

Train Schedule Management System

A full-stack Java web application for browsing and managing train schedules, with role-based login for customers and admins. Built as a team project for Principles of Data Management CS 336 in July 2026.

Overview

Customers can log in and either reserve, view schedules by route and date, delete train schedules. Admins can log in separately to add, update, or remove scheduled trains and view customer accounts.

Tech Stack

- **Backend:** Java (Servlets)
- **Database:** MySQL, accessed via JDBC
- **Frontend:** JSP

Features

- Role-based authentication for customers and admins
- Secure login using parameterized SQL queries (JDBC PreparedStatement) to prevent SQL injection
- Session management to maintain logged-in state across pages
- Schedule search/browsing, booking, admin CRUD on trains.

My Contributions

This was a 3-person team project. I was responsible for:

- Designing and implementing the authentication system, including the database connection layer (ApplicationDB.java) and login logic (LoginServlet.java)
- Writing parameterized SQL queries to authenticate users against the MySQL users table
- Implementing session management for logged-in customers and admins
- Schema design.

Notes

This was a class project focused on core CRUD and authentication functionality rather than production security practices — passwords are not hashed, for example. In a production system, this would be addressed with a proper hashing algorithm (e.g. bcrypt) before storage.